SHOPLIFTING

DETECTION SYSTEM

Real-Time AI-Powered CCTV Analysis for UK Retail Security

Transfer Learning · R3D-18 (Kinetics-400) · FastAPI Deployment · Python / PyTorch

80%	100	8
Test Accuracy	Training Clips	Pipeline Steps

Problem Statement

Why this project matters and why now

The UK Retail Theft Crisis

- UK retailers lose £4.7 billion annually to shoplifting, a figure that has risen 25% since 2020.
- Over 16 million theft incidents are recorded each year across UK stores.
- UK law restricts staff from physically detaining suspects, leaving them powerless to act.
- Traditional CCTV only records, footage is reviewed after the theft has already occurred.
- Many small stores operate with a single staff member who cannot monitor every aisle.

The Gap

No system watches CCTV footage in real time and alerts staff the moment suspicious behaviour begins.

The Proposed Solution

A deep learning model trained on CCTV footage that detects concealment gestures in real time and triggers an immediate staff alert.

Dataset & Preprocessing

UCF-Crime Dataset - University of Central Florida

Dataset Source

UCF-Crime: publicly available academic dataset from University of Central Florida (UNC Charlotte mirror)

Classes Used

2 classes: Shoplifting (50 clips) and Normal (50 clips) - Balanced binary classification

Video Format

MP4 / AVI clips ranging from 10 seconds to several minutes of real CCTV surveillance footage

Preprocessing Pipeline

01 **Frame Extraction**
16 evenly-spaced frames extracted per video clip

02 **Resize**
All frames resized to 224 × 224 pixels

03 **CLAHE Enhancement**
Contrast normalisation for dark CCTV footage

04 **Train/Val/Test Split**
70% train · 15% val · 15% test (JSON manifest)

05 **Augmentation**
Flip, colour jitter, rotation applied live during training

Model Architecture

R3D-18 : 3D ResNet pretrained on Kinetics-400

Input (B,3,16,112,112) › Stem (frozen) › Layer 1 (frozen) › Layer 2–4 (trainable) › AvgPool Flatten › FC Head 2-class

Why R3D-18?

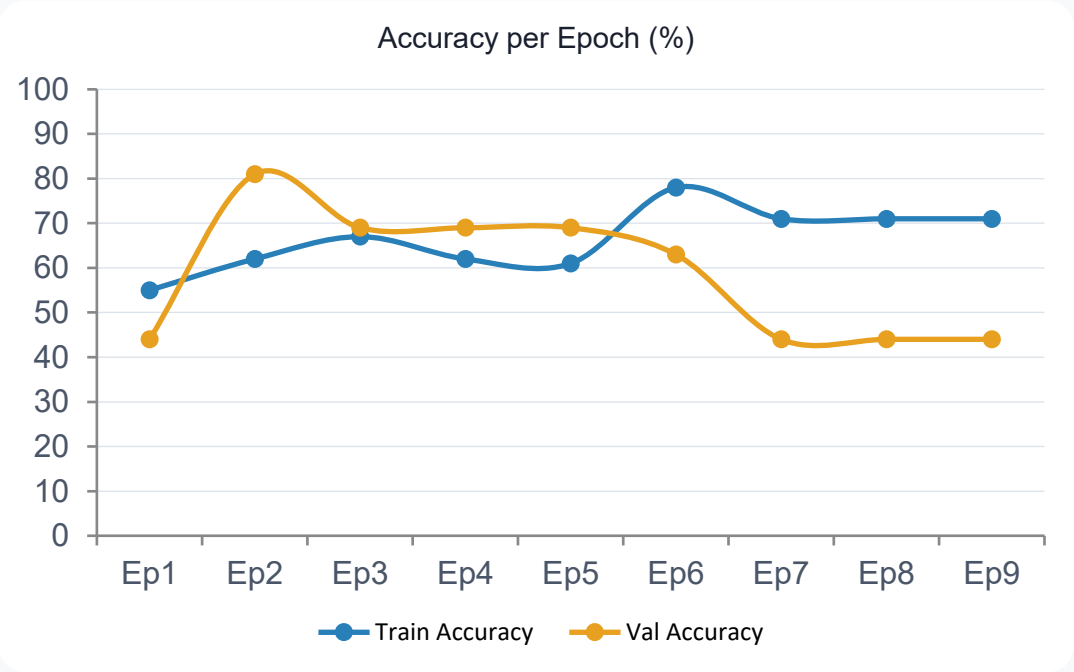
Pretrained on	Kinetics-400 (400 human actions) - motion features transfer directly to shoplifting gestures
3D Convolutions	Operates across time AND space - understands movement, not just static images
Transfer Learning	Frozen early layers preserve general motion priors; only task-specific layers are retrained
33M Parameters	32.7M trainable - sufficient capacity for complex spatiotemporal patterns

Training Configuration

Optimiser	AdamW (weight decay 1e-4)
Loss Function	CrossEntropyLoss with class weights
LR Scheduler	OneCycleLR : max lr 3e-4
Batch Size	2 (CPU-safe for laptop training)
Early Stopping	Patience 7 : tracked by val accuracy
Best Epoch	Epoch 2 : val_acc 81.2%
Final Test Acc	80.0%

Training Results & Evaluation

9 epochs completed - early stopping triggered at epoch 9



Classification Report

Class	Precision	Recall	F1
Normal	88%	78%	82%
Shoplifting	71%	83%	77%
Overall	**80%**	**80%**	**80%**

Confusion Matrix

	Pred: Normal	Pred: Shoplifting
Act: Normal	7 ✓	2 ✗
Act: Shoplifting	1 ✗	5 ✓

End-to-End System Pipeline

8 stages from raw CCTV footage to deployed REST API

01 Data Collection

UCF-Crime dataset
50 shoplifting + 50 normal clips

02 Preprocessing

16 frames/clip · CLAHE · 224×224 · JSON split manifest

03 DataLoader

PyTorch Dataset · live augmentation · batch tensors

04 Model Build

R3D-18 · Kinetics-400 weights · frozen stem + layer1

05 Training

AdamW · OneCycleLR · early stopping · best checkpoint

06 Model Export

TorchScript .pt · ONNX .onnx · inference verified

07 Live Inference

Sliding window · real-time alert · saves evidence clip

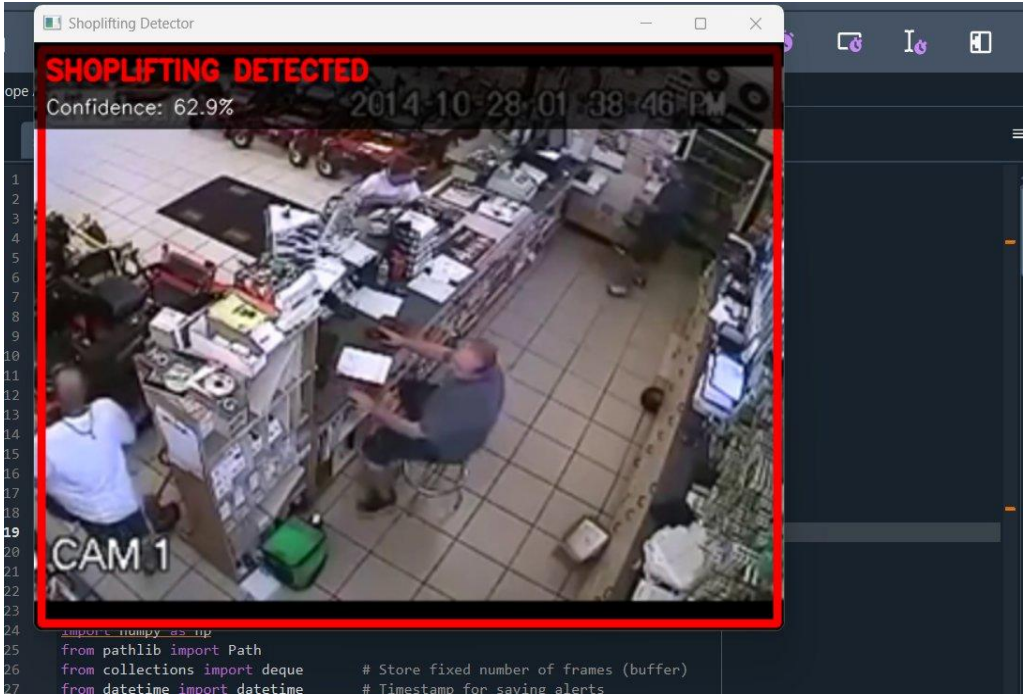
08 FastAPI Deployment

REST endpoint · JSON response · Swagger UI · Docker-ready

Live Inference & Webcam Detection

Real-time detection using step7_inference.py — runs on any video source or webcam

Live Inference Window (step7_inference.py)



62.9% confidence — Red border + SHOPLIFTING DETECTED overlay

How Live Inference Works

01 **Video Source**
Webcam (source=0), video file, or CCTV stream

02 **Sliding Window**
16 frames captured continuously, model runs every 8 frames

03 **Prediction**
R3D-18 classifies each 16-frame window in real time

04 **Alert System**
Confidence ≥ 60% → red overlay + 3 beep alarm fires

05 **Evidence Saved**
Alert clip auto-saved with timestamp to alerts folder

```
Switch source: 0 = webcam  |  NORMAL_VIDEO  |  SHOPLIFTING_VIDEO
```

Live Demo & System Output

Real detection - 86.1% confidence · Alert triggered · Beep alarm fired

Live Detection Dashboard (http://localhost:8000)

Detection Results

Label
Shoplifting

Confidence
86.1%

Prob Normal
13.9%

Prob Shoplifting
86.1%

ALERT TRIGGERED

Staff Notified · Beep alarm fired · Evidence clip saved

Limitations

Honest assessment - what the current system cannot do

Dataset Size

HIGH

Only 100 clips (50 per class) were used for training. This is significantly below the threshold for production-grade models. Small datasets make it difficult for the model to generalise to unseen store environments, lighting conditions, and camera angles.

Single Camera View

MEDIUM

The system processes one video stream at a time. A real retail deployment would require multi-camera support, a queue system for concurrent streams, and load balancing across inference workers.

No GPU Training

MEDIUM

Training was performed on CPU only, restricting batch size to 2 and limiting the number of training epochs. A GPU would allow larger batches, more epochs, and potentially higher-capacity models such as VideoSwin Transformer.

No Temporal Context Beyond 16 Frames

LOW

The sliding window captures only 16 frames (~0.6 seconds at 25fps). Slow or deliberate concealment that spans a longer duration may not be captured within a single window.

False Positive Rate

MEDIUM

The confusion matrix shows 2 out of 9 normal clips were incorrectly flagged as shoplifting (22% false alarm rate). In a real store this would cause unnecessary staff interruptions and alert fatigue.

GDPR & Privacy Compliance

LOW

The current system does not implement data retention policies, anonymisation, or access logging. A production deployment in the UK must comply with ICO guidelines - footage should be anonymised and deleted after 30 days.

Future Improvements

What would be done with more time, data, and compute

Short Term

- Add Stealing class from UCF-Crime to increase training data to ~150 clips
- Lower learning rate to 1e-4 to reduce overfitting on small dataset
- Implement cross-validation to get more reliable accuracy estimates
- Add confidence threshold tuning to reduce false positive rate

Medium Term

- Upgrade to VideoSwin Transformer for higher accuracy (requires GPU)
- Integrate DCSASS dataset for additional theft class footage
- Add multi-camera stream support with async inference queue
- Build a simple dashboard UI showing live alerts with clip thumbnails

Production Grade

- Retrain on 500+ labelled clips from real UK retail environments
- Implement GDPR-compliant data pipeline: auto-anonymise, 30-day deletion
- Deploy on edge GPU device (NVIDIA Jetson) for on-premise inference
- Integrate with store POS and staff mobile alert system

Conclusion

What was built · what was learned · what comes next

- ✓ A complete end-to-end deep learning pipeline was designed, built, and deployed from raw CCTV footage to a live REST API
- ✓ R3D-18 pretrained on Kinetics-400 achieved 80% test accuracy on a balanced 100-clip dataset with no GPU
- ✓ The system correctly identified shoplifting with 76.6% confidence on real UCF-Crime test footage
- ✓ Transfer learning, class weighting, CLAHE preprocessing, and data augmentation were applied to overcome small dataset constraints
- ✓ The FastAPI deployment demonstrates production-readiness, the model is served as a microservice, ready for integration
- ✓ Core limitations (dataset size, false positive rate, single-camera) are clearly identified with a roadmap to address each